

UVM factory

UVM builds most testbench objects through a factory, not with bare new everywhere

Create, type override, instance override

- Registration macros put each class on the factory menu
- Creation asks for a type by name or type handle
- A type override says whenever anyone requests base driver
- An instance override targets one hierarchy path
- If both apply at the same path, instance wins
- Sequences and agents follow the same idea

Browser lab

Digital Design and Verification Platform

HomeTools

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INTERACTIVE TOOL

Factory override sketch

See **who actually gets constructed** when code asks for a type: `type_id::create + type/instance override`. Starter: request `base_driver`, type-override to `error_driver`.

Starter example: code asks for `base_driver`; a type override redirects `create` to `error_driver`.

Load starter example

Challenges 0 / 22 cleared

Quiz: **create:** Preferred way to build a UVM component is...

☐ `type_id::create` through the factory

☐ always raw `new()` with no registration

☐ \$finish inside `build_phase`

☐ only \$readmemh

Show hint

Check

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Core ideas

create

Ask the factory for a type by name — do not new components by hand.

Type override

Everywhere: requested base → alternate derived type.

Inst override

One hierarchical path gets a different type.

Why

Swap error/VIP variants without rewriting the env.

UVM factory lab starter

learn_uvm2017 — uvm factory

Real UVM literacy

```
❯ uel - module05-uvm-factory/examples/navigation

# Track A - Real Unix
# real shell session (Track A - UVM factory literacy)

$ pwd
/mnt/d/proj/designs/Learning/courses/Learn_uvm2017/module05-uvm-factory

$ ls examples/factory-sketch/
overrides.txt

$ cat examples/factory-sketch/overrides.txt
UVM factory sketch (Literacy - overrides)

Registration:
'uvw_component_utils(my_driver)
'uvw_object_utils(my_sequence)

Creation (typical):
my_driver::type_id::create("drv", this); // parent = agent

Vocabulary:
requested type - what the source code asks the factory for
constructed type - what the factory actually builds

Type override (global for that type):
set_type_override_by_type(base_driver::get_type(), error_driver::get_type())
+ every create(base_driver) yields error_driver

Instance override (one path):
set_inst_override_by_type(base_driver::get_type(), error_driver::get_type(),
                          "env_agent.drv")
+ only that path gets error_driver

Precedence at a path:
instance override > type override > no override (identity create)

Common snap targets:
driver / monitor variants (error injection, quiet mode)
sequence types (stress vs smoke)
whole agent (base_agent + vip_agent)

Timing:
set overrides before create - usually early in build phase

Common mistakes:
using 'new' on UVM components (bypasses factory)
setting overrides after create (too late)
forgetting which path an instance override matches

$ head -24 ../../Learn_uvm2017_sv_verilator/tools/uvm-2017/1000.2-2017-1.0/src/base/uvw_factory.svh
//
//
// Copyright 2007-2014 Mentor Graphics Corporation
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// Copyright 2018 Qualcomm, Inc.
// Copyright 2011 Synopsys, Inc.
// Copyright 2007-2018 Cadence Design Systems, Inc.
// Copyright 2011 Verilab
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```

Real shell — factory override sketch

Pitfalls to watch

- Do not bypass the factory with new on UVM components, you lose override hooks
- Do not set overrides after the object is already constructed
- Do not confuse type and instance overrides
- And remember

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, run create on the starter and explain the constructed type without looking
- On real UVM, sketch one type override and one instance override for a driver swap
- When you are ready, take the short quiz, then continue to ConfigDB in the next module